DIGITAL EGYPT PIONEERS INITIATIVE / GRADUATION PROJECT / TEAM 2

WEATHER INSIGHTS & DATA RELIABILITY

Forecast Verification • Travel Season Recommendations
Optional Aviation Route Weather Analysis

01 / COMPARE

Forecasts vs actual readings

02 / RECOMMEND

Best months to visit Egyptian cities

03 / EXPLORE

Weather along a flight corridor

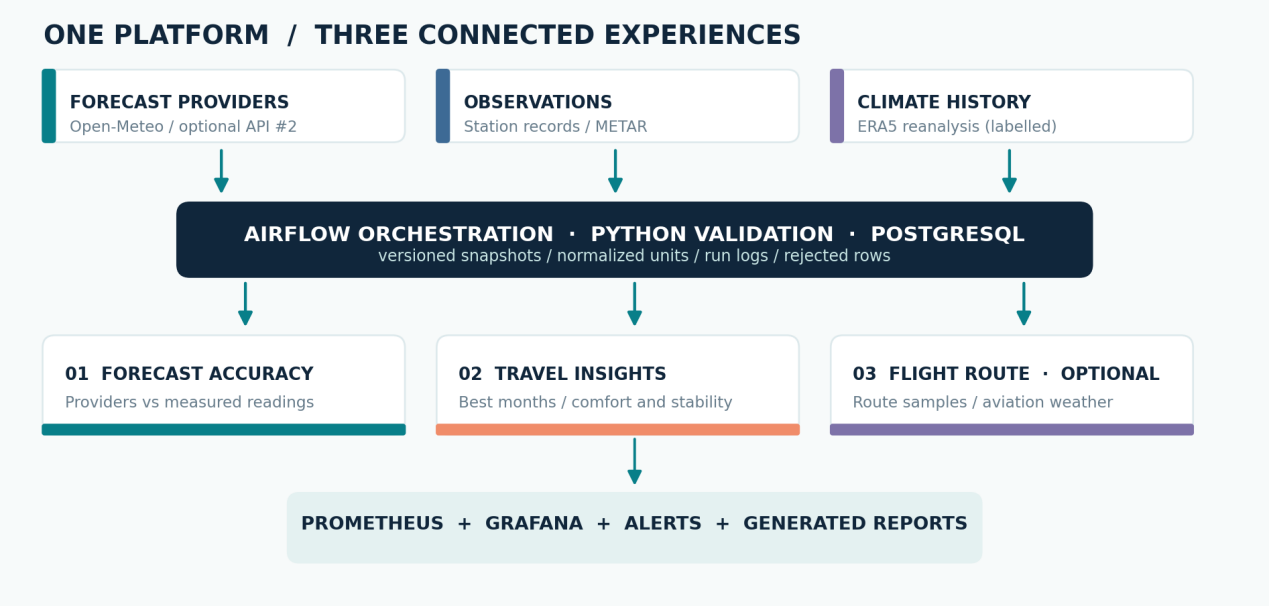


Figure 1. Proposed system at a glance: two core product modules, one optional extension, and shared observability.

Status: proposal and requirements baseline

All charts, scores, alerts and example results in this report are illustrative unless explicitly identified as sourced datasets.

Executive Summary

This graduation project proposes a weather-focused Data Engineering platform that preserves weather forecasts over time, compares them fairly with later observations, and converts historical patterns into explainable travel-season reports. Rather than acting as another weather app, the platform aims to prove data provenance, measurable forecast accuracy, reliable pipelines and defensible recommendations.

Core module A — Forecast Accuracy Lab: collect forecasts from at least two distinct providers (or clearly identified models for the initial prototype), retain the forecast as it was issued, and match each prediction with the corresponding observed weather measurement after the valid hour passes. Compare providers by city, weather variable, and forecast lead time.

Core module B — Travel Weather Intelligence: aggregate multi-year climate history, evaluate configurable comfort and stability criteria, then generate a city/month suitability report. Example query: “Which months are generally best to visit Aswan for comfortable, predictable outdoor weather?” The answer must describe data years, score logic and limitations rather than claim a guaranteed future condition.

Optional module C — Aviation Weather Along a Route: combine an illustrative route and timestamps with waypoint weather and airport METAR/TAF information to demonstrate route-focused weather visualization. This is an educational analysis only, not an operational flight-planning or safety tool.

What makes the graduation project distinctive

A single reproducible pipeline connects forecast snapshots, later verification, climate-aware recommendations, record-level data quality, Apache Airflow orchestration, Prometheus/Grafana observability, and an evidence-based final report.

Document Guide

Three parts: I. Project Planning & Management; II. Literature Review & Evaluation; III. Requirements Gathering & Specification. Source references, acceptance tests and the final demo plan follow the requirements.

Team Members

Student	Student ID	Proposed workstream
Habiba Mohamed Ahmed Ali	21123597	Forecast providers and observation connectors
Sameh Mohamed El-Araby Mohamed	21166667	ETL orchestration and API reliability
Fares Youssef Ali Youssef	21130011	Versioned database model and forecast verification
Mohamed Ramadan Fathy Mohamed	21039161	Monitoring, dashboards and alerts
Mahmoud Abdel-Sabour Bayoumi Sweilem	21179464	Travel scoring, testing and reporting
Alaa Ibrahim El-Saeed El-Hariry	21042691	Team coordination, research and documentation

Roles are suggested based on the updated scope and must be confirmed by the team. The original team list also includes Arabic-name spellings. Contact information is excluded from this public-facing academic draft.

PART I

Project Planning & Management

1. Project Proposal & Problem Statement

Problem: weather providers revise their predictions continuously, while downstream users often compare the latest forecast with the weather reported afterward. This approach can produce misleading accuracy claims because the historical prediction may have been overwritten. At the same time, travel advice is frequently based on informal seasonal assumptions rather than an explicit multi-year methodology. Both workflows are exposed to missing values, mixed units, station/coordinate mismatch, API failures and ambiguous source types.

Objective: build a small but production-oriented weather data platform that (1) validates, versions and monitors forecast/observation data, (2) measures forecast accuracy fairly, and (3) produces explainable monthly travel recommendations, with optional aviation-route enrichment.

2. Project Scope & Prioritization

Priority	Included deliverable	Boundary
MVP / Core	Two forecast sources/models, one observation-source integration, hourly snapshots, forecast matching, error metrics, Airflow, PostgreSQL, dashboards	Start with Cairo and Aswan; expand to 5 pilot cities after validation
MVP / Core	Aswan monthly tourism report using selected historical dataset and 5+ years where available, transparent scoring and a generated PDF	Comfort preferences are configurable; avoid claiming scientific objectivity
Stretch	3rd provider, precipitation and wind verification, automated comparison ranking, additional cities	Only if matched coverage is adequate
Optional	Illustrative route-weather timeline with airport weather and planned route coordinates	Not a real-time pilot aid; no claims to predict actual aircraft path
Excluded	Building an original numerical weather model; operational airline dispatch; guaranteed climate or travel predictions	Outside project resources and remit

3. Example Questions the Platform Answers

- For a date and city: Which provider made the most accurate 24-hour temperature forecast?
- Across the last 30 days: Did Provider A outperform Provider B for Cairo, with equal paired samples?
- For Aswan: Which calendar months had more days meeting our stated outdoor comfort limits?
- For a proposed Cairo–Aswan flight: What weather products could be displayed at departure, arrival and intermediate sampled coordinates?

4. Proposed Product Architecture

The architecture is shown on the cover (Figure 1). Three input families feed a shared ingestion and quality layer: forecast snapshots, station observations, and historical climate/reanalysis. Apache Airflow orchestrates collection; Python checks and PostgreSQL store the reproducible records; Prometheus/Grafana monitor operations. The forecast accuracy and travel recommendation products are core, while flight weather is feature-gated.

5. Collection & Verification Workflow

1. At each scheduled collection, fetch hourly forecasts for selected cities; store request parameters, source/model, forecast_issued_at, forecast_valid_at, lead time and the original response.

2. Fetch observations from a selected station source; record station ID, position, observation time, quality flags, variable definitions and collection time.
3. Normalize units, time zones and variable meaning. Validate nulls, duplicates, schema, bounds, hourly coverage and time alignment; separate invalid from suspicious records.
4. After observed data becomes available, pair by variable and valid time, using a documented maximum distance/time tolerance. Reject incompatible pairs from benchmark calculations.
5. Calculate MAE, RMSE, bias, missing-observation coverage and sample size per provider/city/lead. Generate comparison charts and retain reproducible evaluation runs.
6. Build historical monthly features, apply a documented weighted comfort score, render travel reports, and expose pipeline performance metrics and alerts.

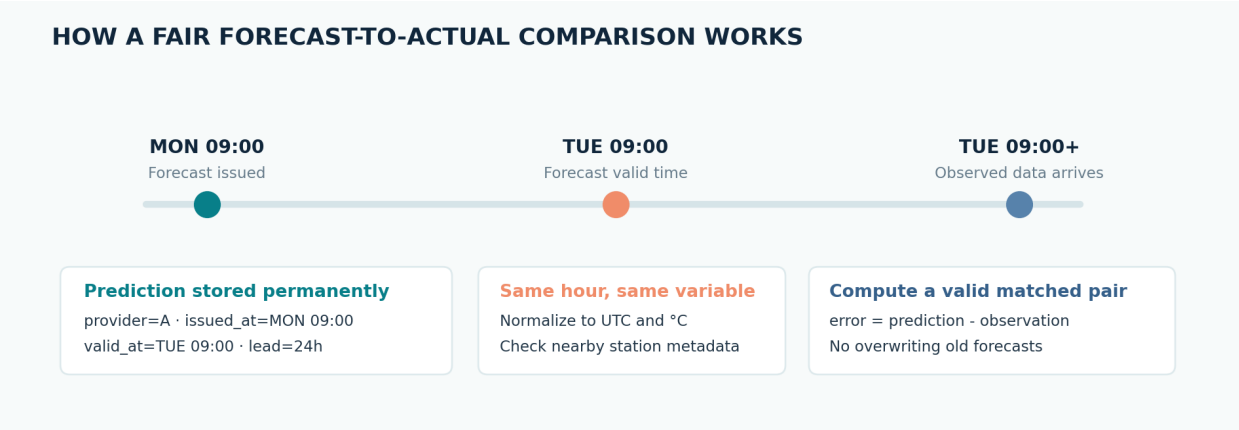


Figure 3. Historical forecast snapshots must be preserved. Joining a newly updated forecast to yesterday's "actual" is not a valid skill comparison.

6. What Counts as "Actual" Weather?

Critical data-science distinction

Station observations (e.g., NOAA Global Hourly or selected METAR stations) are measured reports, subject to coverage and quality limitations. ERA5 and similar reanalysis products are model-informed estimates, useful as a labelled reference proxy but not equivalent to direct station measurements. Forecast models or websites should never be used as their own ground truth.

A reference station may be many kilometres from the target city centre (especially airport METAR). Save station distance and elevation; label comparisons "station-reference" and report match coverage. Never mix temperature measured at different heights or rainfall reporting intervals without transformation.

7. Delivery Plan, Milestones & Team Coordination

Suggested internal schedule: eight working weeks after documentation approval. This is a team planning scenario; the official DEPI final project deadline must be confirmed.

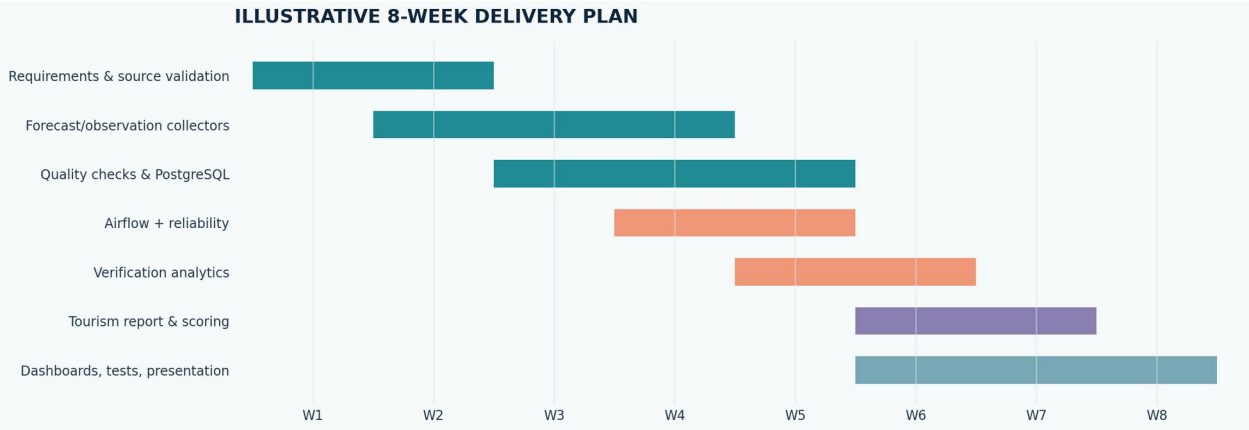


Figure 4. Illustrative Gantt-style roadmap for the core MVP; the aviation route feature is a bonus only after core milestones pass.

Milestone	Target week	Evidence for acceptance
M1: data contracts and source trial	W2	API samples, station availability check and normalized schema
M2: ingestion and immutable snapshots	W4	Versioned forecast rows and observed readings in PostgreSQL
M3: metrics and tourism prototype	W6	Paired MAE chart, Aswan month-ranking prototype and scoring tests
M4: complete observability	W7	Airflow DAG, retried failure scenario, Grafana dashboard and alerts
M5: final demo and documentation	W8	Reproducible Docker setup, report, charts and lecturer demo

8. Collaboration Model & Resource Allocation

GitHub workflow: protect main; use one issue per deliverable; require a pull request, tests and at least one peer review. Each module has an owner and independent reviewer, but every member should participate in integration testing. Weekly milestone review; document API access restrictions and the source-to-field data contract before parallel development.

9. Risk Register & Mitigations

Risk	Priority	Mitigation / contingency
No reliable nearby station data / biased station location	High	Pilot station-available locations first; disclose distance; use ERA5 only as separately labelled proxy
Provider historical forecasts not accessible in plan	High	Collect and persist forecasts prospectively; use prior-run archives where available; choose alternate model instead of fabricating backtests
Unfair comparison from different lead times or variables	High	Immutable issued_at and valid_at fields; same lead window, units and paired samples

Risk	Priority	Mitigation / contingency
Short historical period produces misleading travel ranks	High	Backfill 5–10+ years when permitted; show coverage and sample counts; season recommendations as tendencies
API rate limits / outages	Medium	Backoff, caching, sensible collection cadence and recorded replay fixtures
Extreme weather flags incorrectly rejected	Medium	Physical impossibilities fail hard; plausible extremes are warnings requiring review
Scope explosion caused by aviation module	High	Feature gate; implement only after all core requirements and test cases pass
Incorrect aviation claims	High	Educational route only; no safety certification or real operational dispatch advice

10. Proposed KPIs

Metric	Draft measurable goal	How to demonstrate
Forecast snapshot reproducibility	100% of evaluated predictions retain issued_at + valid_at + provider	Query archived snapshots and rerun same evaluation
Data-quality detection	100% of seeded deterministic invalid fixture cases detected	Automated unit test report
Pair validity / provenance	100% of score rows linked to a reference source and match rules	Audit forecast_match rows
Pipeline idempotency	0 duplicate natural keys after replay	Rerun a backfill interval
Report transparency	Every recommendation displays data period, score inputs and coverage	Review an exported Aswan report
Observability coverage	All scheduled runs logged; last run exposed to dashboard	Failure simulation and Grafana panels
Demo performance	Small demonstration batch completes within agreed local time budget	Timed five-run benchmark

These are internal acceptance targets, not measured results and not official DEPI grading criteria. Forecast accuracy itself is evaluated as a finding (MAE/bias), not promised in advance.

PART II

Literature Review & Related Work

11. Research Questions & Method

This focused technical review uses official documentation and data-source descriptions. It is not a systematic peer-reviewed meteorology review. Its goal is to justify the project's data design and experimental method.

- RQ1: How do archived forecast runs differ from historical climate reanalysis and ground observations?
- RQ2: How should multi-provider forecasts be compared fairly across city, variable, and issuance lead?
- RQ3: How can monthly climate summaries produce transparent travel suitability scores?
- RQ4: What observability capabilities are necessary to make forecast experiments reproducible and dependable?
- RQ5: Which aviation datasets can enrich a route demonstration without making operational-safety claims?

12. Sources & Their Appropriate Uses

Data / technology	What it contributes	Important qualification
Open-Meteo Forecast + Previous Runs [1,2]	Hourly predictions and fixed lead-time historical forecasts	Model-derived predictions; preserve issued/valid times, model identity
Open-Meteo Historical Weather [3]	ERA5/ERA5-Land multiyear monthly features	Reanalysis is not station ground truth
NOAA Global Hourly ISD [4]	Observed hourly station records	Quality controls, variable availability and coverage vary by station
WeatherAPI.com [5]	Independent provider candidate for current/forecast API	Key, access plan, retention and forecast horizon require validation
AviationWeather.gov [6]	METAR airport observations; TAF forecasts	Limited lookback/rate restrictions; airport-centric
OpenSky Network [7]	Historical flight tracks and waypoints	Historical track access is restricted; not future guaranteed routes
Airflow, Prometheus, Grafana [8–10]	Scheduling, operational metrics and alert visualization	Airflow task success does not establish correct data content

13. Accuracy Evaluation Methodology

For each provider p, city c, variable v and lead horizon h, retain only matched pairs (forecast_i, observation_i) satisfying the same reference, units and time tolerance. For N matched samples: MAE = mean(|forecast_i − observation_i|); RMSE = sqrt(mean((forecast_i − observation_i)^2)); bias = mean(forecast_i − observation_i). Report N and coverage alongside every score. Smaller MAE/RMSE is preferable, while bias indicates directional over/underprediction.

A fair ranking requires the same valid hours for every compared provider (paired evaluation). Report stratified results for +24h / +48h / +72h, not a mixture of forecasts with changing horizon. For rain occurrence, use appropriate event-oriented measures rather than forcing temperature MAE onto binary rain events.

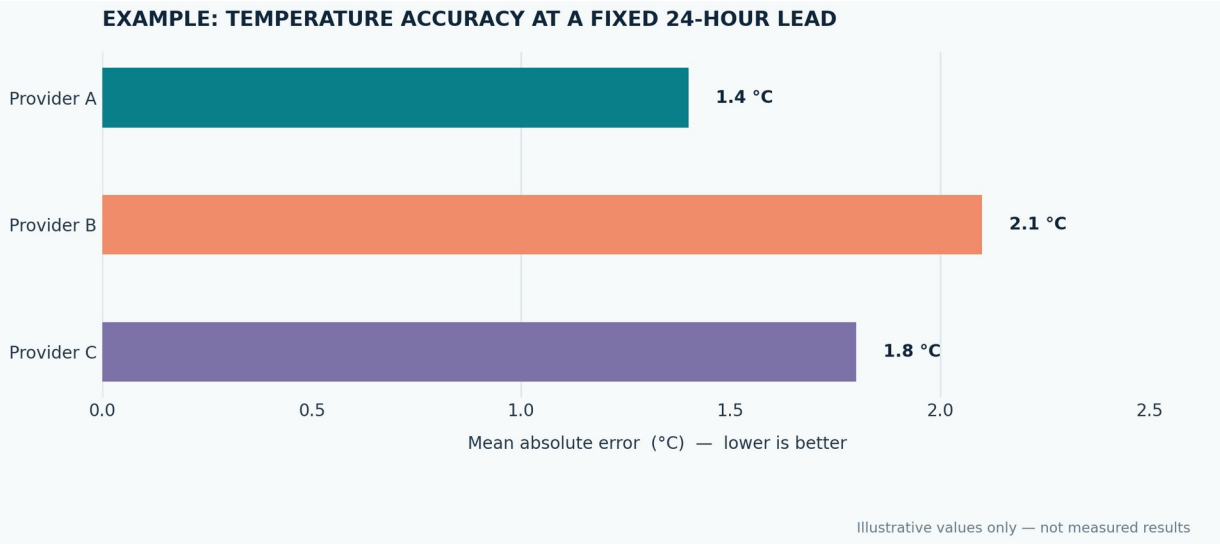


Figure 5. Hypothetical provider MAE comparison; example numbers only. The actual leaderboard must be computed from verified paired station observations.

14. Why Source Independence Matters

Multiple weather websites may reuse identical underlying forecast models. The project should document source organisation, forecast model (when available), update cadence, and data lineage. The MVP may compare GFS/ECMWF/ICON models accessed through an aggregator, but should label that experiment “model comparison,” not claim independent commercial website measurements. A true multi-provider demonstration should integrate a second authorised provider endpoint.

15. Historical Climate & Explainable Tourism Scoring

A “best month” recommendation is a user preference score rather than an objective meteorological fact. Define an outdoor travel profile (e.g., desired apparent temperature, lower rainfall probability, moderate wind, low variability) and compute transparent monthly summaries over multiple years. Record the fraction of available hours/days, medians and percentiles, not only the mean.

Feature	Illustrative definition	Reporting requirement
Thermal comfort	Share of daytime hours within user-selected feels-like temperature band	Display band and number of hours
Precipitation	Share of days with precipitation under a specified threshold	State the threshold and whether reanalysis or observations
Wind comfort	Share of daytime hours below wind threshold	Units, station/model source and coverage
Stability	Relative variability of daily temperature / rapid changes across years	Explain whether stable means low variation or low extreme-event rate
Confidence / coverage	Available records / expected records and years represented	Lower confidence or withhold rating if insufficient

Prototype weights can be temperature comfort 40%, precipitation 20%, wind 15%, stability 20%, coverage 5%; these are configurable choices, not scientific constants. Expose component scores and sensitivity rather than hiding the algorithm. Compare recommendations across nearby months and across alternative comfort profiles.

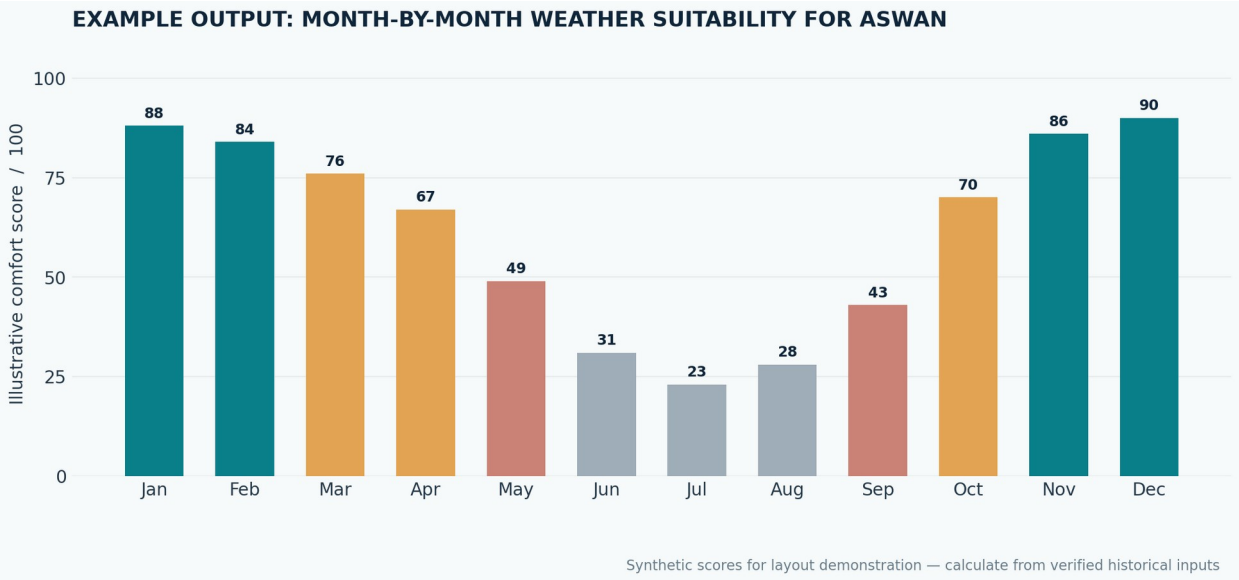


Figure 6. Example Aswan monthly report visual: synthetic suitability scores, not a finding about actual Aswan weather.

16. Related Tools, Engineering Gaps & Contribution

Approach	Strength	Gap our prototype closes
Weather forecast websites	Convenient live predictions	Usually insufficient user-facing access to immutable issue-time snapshots for custom evaluation
Climate history portals	Long time series and trend analysis	Do not directly answer personalized travel suitability with explicit criteria
Standalone Airflow jobs	Reliable scheduling and retry workflows	Task success alone cannot prove forecast accuracy or month-score validity
Prometheus + Grafana	Central metrics and health dashboards	Needs row-level lineage and domain-specific metrics to explain quality
This proposed integrated platform	Provenance + verification + travel insights + observability	Research demo, not a professional meteorological service

17. Lecturer Feedback & Evaluation

Pending external assessment

Lecturer feedback: not yet received. Suggested improvements: to be updated after review. Final grading breakdown: not supplied; do not present an invented rubric as official. Team should attach dated lecturer notes and an updated issue log following feedback.

PART III

Requirements Gathering & Specification

18. Stakeholder Needs & Users

Stakeholder	Need / decision supported	Priority
Data engineer / operator	Schedule ingestion, audit each run, identify stale/missing/failed datasets	Critical
Weather accuracy analyst	Select source/city/variable/lead and inspect fair observed comparisons	Critical
Traveller / trip planner	Explore monthly suitability and underlying climate evidence	High
DEPI supervisor / evaluator	Reproduce system demo and see acceptance evidence	Critical
Team tester	Seed broken records, replay DAG, verify expected alerts and metrics	High
Aviation enthusiast (optional)	View explanatory weather along a proposed corridor	Optional

19. User Stories & Use Cases

Story	User need	Evidence of completion
US-01	As an analyst, I want to compare provider predictions for the same city/date/lead.	Filters + chart with N, MAE, RMSE, bias and coverage
US-02	As a data engineer, I want issued forecasts preserved after API updates.	Old snapshot remains readable after re-ingest
US-03	As a traveller, I want to see recommended Aswan travel months and why.	Monthly scores + thresholds + data years + exported report
US-04	As an operator, I want an alert on pipeline failure or unexpected missing data.	Failed DAG produces traceable alert and status
US-05	As a tester, I want safe backfills and reproducible comparisons.	Repeat run without duplicate rows or changed historic forecast version
US-06	As a student, I want a conceptual flight-corridor weather profile.	Optional route visualization with strong non-operational disclaimer

20. Detailed Use Case: Forecast Benchmark

1. The analyst selects a city, provider pair, observed station reference, variable, date range and fixed +24h forecast horizon.
2. The system finds immutable archived forecast snapshots and station observations with acceptable quality flags.
3. Matching rules require compatible units, the same target valid hour, documented spatial tolerance and comparable issue lead.
4. The system excludes unmatched or low-quality observations, saves reasons and outputs paired-sample count.
5. The report calculates MAE/RMSE/bias and shows time-series discrepancies, coverage and data-source information.

21. Detailed Use Case: Aswan Travel Recommendation

1. The visitor selects Aswan, outdoor activity preference and optionally a different comfort temperature range.
2. The system retrieves precomputed multiyear monthly statistics and checks historical sample coverage.
3. It calculates interpretable factor scores, ranks months, and shows what drove the ranking.
4. It exports a dated PDF report with methodology, years analysed, confidence/coverage and limitations.

22. Functional Requirements

ID	The system shall...
FR-01	Maintain city coordinates, station IDs, source configurations, variable units and licence notes.
FR-02	Collect hourly forecast data periodically from two configured forecast sources or explicitly labelled models.
FR-03	Store immutable forecast issue time, valid time, forecast lead, provider/model and raw retrieval metadata.
FR-04	Ingest reference station observations with source, station coordinates, timestamps and quality flags.
FR-05	Ingest a separate multiyear reanalysis climate dataset for tourism analysis and label its kind.
FR-06	Validate nulls, duplicates, schema/types, supported ranges, reference integrity and expected row counts.
FR-07	Retain invalid/suspicious records with reason, severity, raw payload reference and pipeline run ID.
FR-08	Normalize UTC time and units; prevent mismatched variable definitions from being compared.
FR-09	Match forecasts to later observations using configurable time and station-distance tolerances.
FR-10	Calculate paired-sample MAE, RMSE, bias, N, coverage and lead-specific provider comparisons.
FR-11	Preserve source versions and reproduce historical evaluation results without look-ahead leakage.
FR-12	Compute monthly tourism climate features over a documented historical period.
FR-13	Calculate configurable comfort, rainfall, wind, stability and coverage component scores.
FR-14	Show best-month recommendations with factors, data period, sample count and limitations.
FR-15	Export a dated city/month travel recommendation report in PDF.
FR-16	Schedule periodic ingestion and batch analysis with Apache Airflow and bounded retries.
FR-17	Use idempotent upserts/replay-safe pipeline processing and record success/failure status.
FR-18	Expose pipeline runtime, freshness, records ingested/rejected, match coverage and quality-failure metrics.

ID	The system shall...
FR-19	Display Grafana monitoring panels and trigger alerts on failed/stale runs or excessive invalid records.
FR-20	[Optional] Accept illustrative flight route waypoints and estimated timestamps.
FR-21	[Optional] Attach aviation weather or waypoint model fields while displaying a non-operational disclaimer.

23. Non-functional Requirements

ID	Quality requirement / target (proposed)	Verification method
NFR-01	Reproducible: accuracy results use immutable forecasts and versioned evaluation queries.	Run same historical benchmark twice and compare outputs
NFR-02	Reliability: replay adds no duplicate canonical rows.	Re-ingest identical window twice
NFR-03	Data integrity: missing station coverage is never silently treated as observed weather.	Inspect missing/matched classification
NFR-04	Performance: small demo backfill finishes within a locally agreed budget.	Benchmark 5 runs in documented hardware
NFR-05	Security: API tokens and DB credentials kept out of the Git repository.	Secret scan, .env.example and code review
NFR-06	Observability: run results and rejection counts exposed within 2 minutes of run completion.	Check Grafana fresh-run panel
NFR-07	Usability: source, data years, scoring thresholds and sample counts visible on reports.	5-minute evaluator walkthrough
NFR-08	Portability: core services run through documented Docker Compose.	Setup on clean development machine
NFR-09	Ethics: aviation feature has non-operational notice and no safety claims.	Review screenshots and exported reports

24. Data Model & Record Provenance

Proposed entity	Essential fields / role
providers	id, name, provider_kind, model, units, licence, cadence
locations / weather_stations	coordinates, elevation, station_id, distance from city, reference metadata
forecast_snapshots	provider_id, model, city_id, issued_at_utc, valid_at_utc, lead_hours, variable, value, raw_batch_id
observations	station_id, observed_at_utc, variable, value, QC flags, retrieved_at_utc
forecast_matches	forecast_id, observation_id, match_rule_version, spatial_gap_km, time_gap_min, error
historical_climate_hours	city_id, valid_at_utc, reanalysis_dataset, variable, value, period/version

Proposed entity	Essential fields / role
monthly_city_features	city_id, month, year_range, comfort_pct, rain_score, wind_score, stability, coverage
travel_reports	city_id, preferences_json, score_version, generated_at, input_version, PDF path
pipeline_runs / rejected_rows	run_id, task, timestamps, status, counts, error rule and payload pointer
flight_waypoints (optional)	flight_demo_id, lat/lon, expected_at_utc, altitude, weather_product_ref

Suggested uniqueness: forecasts by (provider/model, city, issue UTC, valid UTC, variable); observations by (station, observation UTC, variable, QC edition); month features by (city, month, dataset version, scoring version). Avoid destructive overwrites of historic runs.

25. Output & Visual Report Concepts

The final demonstration should be understandable to non-technical lecturers in a few minutes. Use two audience-facing reports supported by one internal operations dashboard.

Forecast Verification Dashboard

The comparison graphic in Figure 5 previews a source leaderboard. The live verification page will also show a forecast-versus-station time series, filters for city / provider / variable / +24/+48/+72h lead, sample counts, observation quality flags and downloadable evaluation data.

Travel Planning Report — Aswan

The chart in Figure 6 previews monthly ranking. The final report will add component-score explanations, analysed years, historical coverage, monthly distributions, report-generation date and a confidence caveat. Example wording: “Month X scores highest under the selected daytime comfort preferences and available historical data.”

Optional Route Weather Explorer

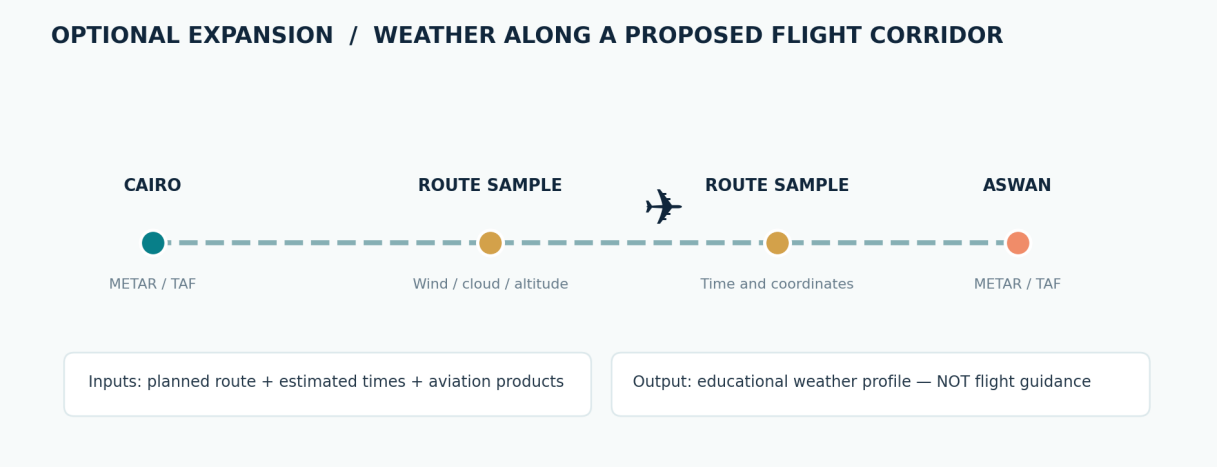


Figure 7. Conceptual planned Cairo–Aswan weather corridor, not an actual flight path or aviation safety product.

A future prototype could sample a planned great-circle route at fixed progress intervals and interpolate estimated times, then display suitable forecast-model fields along the route plus origin/destination METAR/TAF. A real aircraft route and altitude profile require external flight-plan data; upper-air weather may require specialized datasets. Ground-level city forecasts are not sufficient to infer en-route flight conditions.

26. Acceptance Test Scenarios

Test	Fixture / user action	Expected evidence
AT-01	Provider revises forecast for an existing valid hour	Earlier issued snapshot still present; lead calculations unchanged
AT-02	Join a +24h forecast with a +72h forecast	Comparison refused or segregated by horizon
AT-03	Station observation missing / too distant	Marked unmatched, counted in coverage, excluded from MAE
AT-04	Humidity value = 150% / duplicate forecast key	Invalid or duplicate is recorded with explicit reason
AT-05	Evaluate two providers on unequal coverage	Use common intersection of valid paired hours
AT-06	Repeat identical ingestion window	No additional canonical duplicates
AT-07	Generate Aswan report with missing historical years	Coverage warning and score confidence reduced or withheld
AT-08	Simulate temporary API failure	Airflow retry; logged recovery or final failure alert

Test	Fixture / user action	Expected evidence
AT-09	Export monthly trip report	Report contains city, methodology, weights, time period, provenance
AT-10	Open optional aviation report	“Not for operational flight planning” notice visible

27. Demo Walkthrough & Expected Deliverables

1. Open Airflow and trigger the hourly forecast job for Cairo and Aswan; show raw response metadata and immutable issue-time forecast records.
2. Run data-quality cases (nulls, impossible humidity, duplicate rows) and inspect rejected record reasons.
3. Load a verified sample observation batch; match past +24h forecasts; demonstrate the provider comparison and honest sample counts.
4. Show the Aswan monthly recommendation report with configurable comfort assumptions, coverage and generated PDF.
5. Temporarily fail an API request; show retries, Grafana quality/freshness metrics and recovery without duplicates.
6. If the core deliverables are complete, show the educational planned-flight weather corridor as a brief optional extension.

28. Decisions Requiring Team / Lecturer Approval

- Which independent provider(s) can we access legally and within the budget? Confirm API keys, historical access, terms and daily forecast horizon.
- Which real station(s) sufficiently represent each Egyptian city? Inspect observation availability and station distances before promising city-specific measured-accuracy results.
- Will accuracy initially target temperature only at +24h, and then expand to wind/precipitation and longer leads?
- How many climate years will be analysed? Which tourism comfort profile and weights should the supervisor approve?
- What is the actual software-delivery deadline? Eight weeks is a proposed plan, not an official DEPI milestone.
- What is the official assessment rubric, and what lecturer feedback is available after reviewing this draft?

29. Final Delivery Checklist

- Approved architecture, scope boundaries and assigned team roles
- Source licences, API access proof and data dictionaries
- Versioned forecast snapshots + observation matching + verifiable accuracy charts
- Historical tourism scoring with explainable Aswan report and data coverage
- ETL DAG, rejected rows, replay tests and run logs
- Prometheus metrics, Grafana dashboards and alert evidence
- Automated tests, sample fixtures, setup guide and recorded demonstration
- Project plan, literature review, requirements and citations revised after lecturer feedback

SOURCE REGISTER

References & Official Documentation

Official API documentation and source descriptions used to define the proposal and its data-science safeguards. Links below should be revisited when the integrations are implemented.

[1] Open-Meteo, Previous Runs API — https://open-meteo.com/en/docs/previous-runs-api

[2] Open-Meteo, Single Runs API — https://open-meteo.com/en/docs/single-runs-api

[3] Open-Meteo, Historical Weather API — https://open-meteo.com/en/docs/historical-weather-api

[4] NOAA NCEI, Global Hourly (ISD) — https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database

[5] WeatherAPI.com, API documentation & plan access — https://www.weatherapi.com/docs/ | https://www.weatherapi.com/pricing.aspx

[6] NOAA Aviation Weather Center, Data API — https://aviationweather.gov/data/api/

[7] OpenSky Network, REST API documentation — https://openskynetwork.github.io/opensky-api/rest.html

[8] Apache Airflow, DAG Concepts — https://airflow.apache.org/docs/apache-airflow/stable/core-concepts/dags.html

[9] Prometheus, Getting Started — https://prometheus.io/docs/introduction/first_steps/

[10] Grafana Documentation — https://grafana.com/docs/grafana/latest/

[11] Open-Meteo, Historical Forecast API — https://open-meteo.com/en/docs/historical-forecast-api

[12] Open-Meteo, Terms & Licences — https://open-meteo.com/en/terms

Before implementation, reconfirm API plans, source coverage and data licences. Bibliographic sources inform the methodology; no external numerical result is represented as a result of this unimplemented project.